

EDUCATION

Georgia Institute of Technology

Ph.D. in Aerospace Engineering, Expected August 2027

Atlanta, GA, USA

Aug. 2024 – Aug. 2027 (Expected)

– Dissertation: Optimization of Sustainable Space Logistics Systems

– Advisor: Prof. Koki Ho

Korea Advanced Institute of Science and Technology (KAIST)

Master of Science in Aerospace Engineering

Daejeon, Republic of Korea

Aug. 2022 – Aug. 2024

– Thesis: Optimal planetary surface routing considering synergy value system and heterogeneous agents

– Advisor: Prof. Jaemyung Ahn

Korea Advanced Institute of Science and Technology (KAIST)

Bachelor of Science in Aerospace Engineering

Daejeon, Republic of Korea

Mar. 2015 – Feb. 2019

Bachelor of Science in Physics (Double Major)

PROFESSIONAL AND RESEARCH EXPERIENCE

Georgia Institute of Technology

Graduate Research Assistant, Daniel Guggenheim School of Aerospace Engineering

Atlanta, GA, USA

Aug. 2024 – Present

Korea Advanced Institute of Science and Technology

Graduate Research Assistant, Department of Aerospace Engineering

Daejeon, Republic of Korea

Aug. 2022 – Aug. 2024

Agency for Defense Development (ADD)

Researcher, Aerospace Technology Research Institute

Daejeon, Republic of Korea

Jun. 2019 - May. 2022

Research Officer for National Defense, Republic of Korea Air Force

PUBLICATIONS

Journal Articles

- [J1] **E. Choi**, J. Kim, J. Kim, E.S. Suh, J. Ahn, “Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution,” *Journal of Mechanical Design*, 2026 (Accepted)
- [J2] **E. Choi**, J. Lee, and J. Ahn, “Optimal Split-Delivery Lunar Campaign with Resonance Orbits and Hybrid Propulsion,” *Journal of Spacecraft and Rockets*, 2026 (Accepted)
- [J3] **E. Choi** and J. Ahn, “Optimal Routing for In-Situ Resource Utilization-Based Planetary Surface Explorations,” *Journal of Aerospace Information Systems*, 2026 (Accepted)
- [J4] **E. Choi** and K. Ho, “Integrated Routing and Trajectory Design for Satellite Servicing,” *Journal of Spacecraft and Rockets*, 2026, pp. 1–15
- [J5] B. Gwon*, **E. Choi***, J. Lee, and J. Ahn, “Adaptive Planning for Multiple UAVs with In-Flight Refueling,” *International Journal of Aeronautical & Space Sciences*, 2026, pp. 1–19
*These authors contributed equally.
- [J6] J. Choi, J. Lee, **E. Choi** and J. Ahn, “Sequential Vertiport Location Optimization Considering Uncertain Social Benefit,” *Journal of Aerospace Information Systems*, Vol. 23, No. 8, 2026, pp. 737–752
- [J7] **E. Choi** and K. Ho, “Orbital Depot Location Optimization for Satellite Constellation Servicing with Low-Thrust Transfers,” *Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 4–13.
- [J8] C. Moon, **E. Choi**, and J. Ahn, “Aerial-Ground Collaborative Routing with Uncertain Exploration Value,” *International Journal of Aeronautical & Space Sciences*, Vol. 26, 2025, pp. 2817–2841.
- [J9] J. Kim, **E. Choi**, J. Ahn, E.S. Suh, J. Kim, and D.G. Lim, “Mathematical Programming-Based Design Structure Matrix Clustering for Modular Architecture Design,” *Journal of Mechanical Design*, Vol. 147, No. 10, 2025, 101401.
- [J10] **E. Choi** and J. Ahn, “Optimal Multi-agent Grouping and Routing with Cooperative Tasks,” *Journal of Aerospace Information Systems*, Vol. 22, No. 3, 2025, pp. 163–176.
- [J11] **E. Choi** and W. Chang, “Multi-Agent Task Allocation with Inter-Agent Distance Constraints,” *Journal of Aerospace Information Systems*, Vol. 21, No. 2, 2024, pp. 168–177.
- [J12] **E. Choi** and J. Ahn, “Optimal Planetary Surface Exploration with Heterogeneous Agents,” *Journal of Spacecraft and Rockets*, Vol. 60, No. 4, 2023, pp. 1343–1354.
- [J13] **E. Choi** and J. Ahn, “Optimal Planetary Surface Routing Considering Exploration Values and Synergies,” *Journal of Aerospace Information Systems*, Vol. 19, No. 6, 2022, pp. 406–420.

Working Papers

- [P1] B. Gwon, **E. Choi**, and J. Ahn, “An Integer Linear Programming-Based Framework for Satellite Coverage Optimization Problem Under Operational Constraints,” *International Journal of Aeronautical & Space Sciences*, (Major revision)

Conference Papers

[C1] **E. Choi** and K. Ho, “Heterogeneous On-Orbit Servicing Architecture with Split Deliveries,” *49th Rocky Mountain AAS GN&C Conference* (Submitted).

[C2] **E. Choi** and K. Ho, “In-Space Recycling Logistics Optimization for Circular Space Economy,” *2027 AIAA SciTech Forum*, (Forthcoming).

[C3] T. Noro, **E. Choi**, S. Han, M. Isaji, and K. Ho, “Optimal Scheduling of ADR Missions in the Near-GEO Region Based on a Competing-Risk Failure Model,” *The 70th Space Sciences and Technology Conference, November 2026* (Forthcoming).

[C4] T. Noro, K. Ho, **E. Choi**, S. Han, and M. Isaji, “Low-Thrust Trajectory Optimization via Sequential Convex Programming for On-Orbit Servicing in GEO,” *The Japan Society for Aeronautical and Space Sciences 57th Annual Meeting*, April 2026.

[C5] **E. Choi** and K. Ho, “Optimal Routing and Trajectory Planning for Asteroid Mining with Partial In-Situ Resource Utilization,” *2026 AIAA SciTech Forum*, January 2026.

[C6] **E. Choi** and J. Ahn, “Optimal Routing Problem with Precedence Constraints for Surface Exploration,” *2024 AIAA SciTech Forum*, January 2024.

[C7] J. Choi, J. Lee, **E. Choi** and J. Ahn, “Optimizing UAM Vertiport Location in Jeju Island: Incorporating Noise for Public Utility,” *2023 Asia-Pacific International Symposium on Aerospace Technology*, October 2023.

[C8] **E. Choi** and J. Ahn, “Optimal Short-Term Scheduling for Space Telescope in Low-Earth Orbit,” *2023 AAS/AIAA Astrodynamics Specialist Conference*, August 2023.

[C9] **E. Choi**, D. Han, and W. Chang, “Decentralized Task Allocation for Multiple UAVs with Iterative Cost Reduction,” *KSAS 2023 Spring Conference*, pp. 60–61, April 2023.

[C10] J. Ahn, **E. Choi** and D.U. Lee, “Application of routing problems to space exploration missions,” *2023 AIAA SciTech Forum*, January 2023.

[C11] W. Chang and **E. Choi**, “Trade-off Study on Multi-UAV Autonomous Mission Planning Algorithms,” *KSAS 2022 Fall Conference*, pp. 1736–1737, 2022.

[C12] **E. Choi** and W. Chang, “Multi-Agent Decentralized Task Allocation Algorithms with Fuel Constraints,” *KSAS 2021 Fall Conference*, pp. 432–433, 2021.

Media Articles

[M1] **E. Choi** and K. Ho, “SpaceX aims to use rockets to quickly transport cargo across the Earth and into orbit – moves that would expand the global cargo system,” *The Conversation*, 2026.

PATENTS AND SOFTWARE DISCLOSURES

1. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, “PosbasedOOSnearGSO.jl,” Software Disclosure, Invention ID 2026-248, 2025.

2. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, “LowthrustSCP.jl,” Software Disclosure, Invention ID 2026-247, 2025.

3. D. Han, W. Jang, **E. Choi**, “Method and apparatus for determining the path of an unmanned vehicle using a distributed network,” *KR Patent*, 10-2946314, 2026.

4. **E. Choi**, K. Ho, and Y. Shimane, “Continuous Location Routing Problem for Orbital Depot (CLRPOD) v1.0.0,” Georgia Tech Software Disclosure, Invention ID 2026-018, 2025.
– <https://github.com/EuihyeonChoi/CLRPOD>

SCHOLARSHIPS AND AWARDS

• Global Korea Scholarship Program for Study Abroad (Ph.D.) by Korean Government Annually \$40,000 and Airfare	Fall 2024 – Summer 2026
• Finalist in 2023 Space Grand Challenge by KAIST, Hanwha Aerospace, and Hanwha Systems	Nov. 2023
• KAIST Scholarship Annually \$15,000 (Full-Ride)	Fall 2022 – Spring 2024
• ADD President’s Award: The Best Research Proposal Subject: Battery Dumping System to Improve Drone’s Endurance	Dec. 2019
• Degree of Bachelor of Science in Cum Laude	Feb. 2019
• Finalist in NASA’s RASC-AL Competition Theme: Design Lunar Polar Sample Return Architecture	Cocoa Beach, FL, USA Aug. 2017 – Jun. 2018
• National Natural Science and Engineering Scholarship Total \$20,000 (Full-Ride)	Spring 2017 – Fall 2018
• The Boeing Company Scholarship Total \$3,500	Spring 2016 – Fall 2017
• National Scholarship for Undergraduate Study Total \$13,500 (Full-Tuition)	Spring 2015 – Fall 2016

TECHNICAL JOURNAL OR CONFERENCE REFEREE ACTIVITIES

- Reviewer, *AIAA Journal of Aerospace Information Systems*, 2025, 2026
- Reviewer, *International Journal of Aeronautical and Space Sciences*, 2026
- Reviewer, *Cluster Computing*, 2026
- Reviewer, *AIAA SciTech*, 2025, 2026
- Reviewer, *AIAA Journal of Guidance, Control, and Dynamics*, 2025
- Reviewer, *The Journal of Supercomputing*, 2025, 2026

TEACHING, MENTORING, AND SERVICE

- **Graduate Teaching Assistant** Georgia Institute of Technology
AE 4532 Spacecraft Flight Dynamics Fall. 2026
- **Research Mentor (External)** University of Colorado Boulder
Mentoring a Master’s student on active debris maneuver and on-orbit servicing research Apr. 2026 – Present
- **Undergraduate Research Mentor** Georgia Tech
Guided the implementation of orbital routing problems and paper replication for space logistics Spring 2026 – Present
- **Invited Speaker** ADD
Delivered an invited talk for the 6th Research Officer for National Defense Sep. 2022
- **Selected Mentor** ADD
Mentoring for the 4th Research Officer for National Defense Oct. 2020 – Jan. 2021

PROJECTS

- Innovative Decision-Making Frameworks for Next-Generation Space Systems Mitsubishi Electric Corporation, Georgia Tech
Oct. 2025 – Aug. 2027
- Orbital Staging of Assets (Phase II) Air Force STTR (Subaward from ThinkOrbital), Georgia Tech
Oct. 2025 – Aug. 2026
- ATLAS: Advancing Technologies for Logistics Architectures in Space AFRL/AFOSR, Georgia Tech
Aug. 2024 – Sep. 2026
- Space Logistics Model for Temporary Deployment of Earth Imaging Satellites ThinkOrbital, Georgia Tech
Feb. 2025 – Apr. 2025
- Mission Planning for Swarm of Tiny Satellites LIG Nex1, KAIST
Oct. 2023 – Jul. 2024
- Tactical Swarm Unmanned Aerial Vehicle Mission Planning and Autonomous Mission Replanning Technology LIG Nex1, KAIST
Oct. 2022 – Jul. 2024
- Modular Architecture Engineering: Extension Considering Practicality Hyundai, KAIST
Sep. 2022 – Jul. 2024
- Routing Problems Considering Value System for Exploration Missions KAIST
Sep. 2022 – Jul. 2024
- Low Observable Wingman UAV System ADD
Nov. 2021 – May. 2022
- Real-Time Decentralized Task Allocation for Multiple and Heterogeneous Unmanned Aerial Systems AFRL, ADD
Aug. 2021 – May. 2022
- Autonomous Navigation and Mission Management Technology ADD
Dec. 2019 – Aug. 2020